

less mysterious than Ext for nonfinitely generated groups, as it commutes with direct limits, according to an exercise for §3.A.

Exercises

1. Given maps $f_i: X_i \rightarrow X_{i+1}$ for integers $i < 0$, show that the ‘reverse mapping telescope’ obtained by glueing together the mapping cylinders of the f_i ’s in the obvious way deformation retracts onto X_0 . Similarly, if maps $f_i: X_i \rightarrow X_{i+1}$ are given for all $i \in \mathbb{Z}$, show that the resulting ‘double mapping telescope’ deformation retracts onto any of the ordinary mapping telescopes contained in it, the union of the mapping cylinders of the f_i ’s for i greater than a given number n .
2. Show that $\varprojlim^1 G_i = 0$ if the sequence $\cdots \rightarrow G_2 \xrightarrow{\alpha_2} G_1 \xrightarrow{\alpha_1} G_0$ satisfies the *Mittag-Leffler condition* that for each i the images of the maps $G_{i+n} \rightarrow G_i$ are independent of n for sufficiently large n .
3. Show that $\text{Ext}(A, \mathbb{Q}) = 0$ for all A . [Consider the homology with $\mathbb{Q}$ coefficients of a Moore space $M(A, n)$.]
4. An abelian group G is defined to be *divisible* if the map $G \xrightarrow{n} G$, $g \mapsto ng$, is surjective for all $n > 1$. Show that a group is divisible iff it is a quotient of a direct sum of $\mathbb{Q}$ ’s. Deduce from the previous problem that if G is divisible then $\text{Ext}(A, G) = 0$ for all A .
5. Show that $\text{Ext}(A, \mathbb{Z})$ is isomorphic to the cokernel of $\text{Hom}(A, \mathbb{Q}) \rightarrow \text{Hom}(A, \mathbb{Q}/\mathbb{Z})$, the map induced by the quotient map $\mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z}$. Use this to get another proof that $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}) \approx \hat{\mathbb{Z}}_p$ for p prime.
6. Show that $\text{Ext}(\mathbb{Z}_{p^\infty}, \mathbb{Z}_p) \approx \mathbb{Z}_p$.
7. Show that for a short exact sequence of abelian groups $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, a Moore space $M(C, n)$ can be realized as a quotient $M(B, n)/M(A, n)$. Applying the long exact sequence of cohomology for the pair $(M(B, n), M(A, n))$ with any coefficient group G , deduce an exact sequence

$$0 \rightarrow \text{Hom}(C, G) \rightarrow \text{Hom}(B, G) \rightarrow \text{Hom}(A, G) \rightarrow \text{Ext}(C, G) \rightarrow \text{Ext}(B, G) \rightarrow \text{Ext}(A, G) \rightarrow 0$$
8. Show that for a Moore space $M(G, n)$ the Bockstein long exact sequence in cohomology associated to the short exact sequence of coefficient groups $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ reduces to an exact sequence

$$0 \rightarrow \text{Hom}(G, A) \rightarrow \text{Hom}(G, B) \rightarrow \text{Hom}(G, C) \rightarrow \text{Ext}(G, A) \rightarrow \text{Ext}(G, B) \rightarrow \text{Ext}(G, C) \rightarrow 0$$
9. For an abelian group A let $p: A \rightarrow A$ be multiplication by p , and let ${}_pA = \text{Ker } p$, $pA = \text{Im } p$, and $A_p = \text{Coker } p$ as in the proof of Proposition 3F.12. Show that the six-term exact sequences involving $\text{Hom}(-, \mathbb{Z})$ and $\text{Ext}(-, \mathbb{Z})$ associated to the short exact sequences $0 \rightarrow {}_pA \rightarrow A \rightarrow pA \rightarrow 0$ and $0 \rightarrow pA \rightarrow A \rightarrow A_p \rightarrow 0$ can be spliced together to yield the exact sequence across the top of the following diagram

$$\begin{array}{ccccccc}
 \mathrm{Hom}(pA, \mathbb{Z}) & \longrightarrow & \mathrm{Ext}(A_p, \mathbb{Z}) & \longrightarrow & \mathrm{Ext}(A, \mathbb{Z}) & \xrightarrow{p} & \mathrm{Ext}(A, \mathbb{Z}) \longrightarrow \mathrm{Ext}(pA, \mathbb{Z}) \longrightarrow 0 \\
 & & & & \searrow & \nearrow & \\
 & & & & \mathrm{Ext}(pA, \mathbb{Z}) & & \\
 & & & & \nearrow & \searrow & \\
 0 \longrightarrow \mathrm{Hom}(pA, \mathbb{Z}) & \xrightarrow{\approx} & \mathrm{Hom}(A, \mathbb{Z}) & \longrightarrow & 0 & & 0
 \end{array}$$

where the map labeled ‘ p ’ is multiplication by p . Use this to show:

- (a) $\mathrm{Ext}(A, \mathbb{Z})$ is divisible iff A is torsionfree.
- (b) $\mathrm{Ext}(A, \mathbb{Z})$ is torsionfree if A is divisible, and the converse holds if $\mathrm{Hom}(A, \mathbb{Z}) = 0$.

3.G Transfer Homomorphisms

There is a simple construction called ‘transfer’ that provides very useful information about homology and cohomology of finite-sheeted covering spaces. After giving the definition and proving a few elementary properties, we will use the transfer in the construction of a number of spaces whose $\mathbb{Z}_p$ cohomology is a polynomial ring.

Let $\pi: \tilde{X} \rightarrow X$ be an n -sheeted covering space, for some finite n . In addition to the induced map on singular chains $\pi_\#: C_k(\tilde{X}) \rightarrow C_k(X)$ there is also a homomorphism in the opposite direction $\tau: C_k(X) \rightarrow C_k(\tilde{X})$ which assigns to a singular simplex $\sigma: \Delta^k \rightarrow X$ the sum of the n distinct lifts $\tilde{\sigma}: \Delta^k \rightarrow \tilde{X}$. This is obviously a chain map, commuting with boundary homomorphisms, so it induces **transfer homomorphisms** $\tau_*: H_k(X; G) \rightarrow H_k(\tilde{X}; G)$ and $\tau^*: H^k(\tilde{X}; G) \rightarrow H^k(X; G)$ for any coefficient group G . We focus on cohomology in what follows, but similar statements hold for homology as well.

The composition $\pi_\# \tau$ is clearly multiplication by n , hence $\tau^* \pi^* = n$. This has the consequence that the kernel of $\pi^*: H^k(X; G) \rightarrow H^k(\tilde{X}; G)$ consists of torsion elements of order dividing n , since $\pi^*(\alpha) = 0$ implies $\tau^* \pi^*(\alpha) = n\alpha = 0$. Thus the cohomology of $\tilde{X}$ must be ‘larger’ than that of X except possibly for torsion of order dividing n . This can be a genuine exception as one sees from the examples of S^m covering $\mathbb{R}P^m$ and lens spaces. More generally, if $S^m \rightarrow X$ is any n -sheeted covering space, then the relation $\tau^* \pi^* = n$ implies that $\tilde{H}^*(X; \mathbb{Z})$ consists entirely of torsion elements of order dividing n , apart from a possible $\mathbb{Z}$ in dimension m . (Since X is a closed manifold, its homology groups are finitely generated by Corollaries A.8 and A.9 in the Appendix.)

By studying the other composition $\pi^* \tau^*$ we will prove:

Proposition 3G.1. *Let $\pi: \tilde{X} \rightarrow X$ be an n -sheeted covering space defined by an action of a group Γ on $\tilde{X}$. Then with coefficients in a field F whose characteristic is 0 or a prime not dividing n , the map $\pi^*: H^k(X; F) \rightarrow H^k(\tilde{X}; F)$ is injective with image the subgroup $H^*(\tilde{X}; F)^\Gamma$ consisting of classes α such that $\gamma^*(\alpha) = \alpha$ for all $\gamma \in \Gamma$.*